

Sports betting arbitrage

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Introduction

This is a simple walkthrough of sports betting arbitrage.

Example

Team A is playing against team B. On website 1, A has a 75% chance of winning. On website 2, A has a 25% chance of winning. Recall that when we place a bet of size S on an outcome with probability p , we are risking S to win $\frac{1-p}{p} \cdot S$. (Exercise: verify that this bet has expected value 0.) Thus, notice that when we bet 100 on B on website 1, and 100 on A on website 2, we will end up with $+300 - 100 = +200$ no matter what the outcome! (Exercise: verify this.)

General case

Let p be the probability that A wins on website 1 and q be the probability that A wins on website 2. For an arbitrage to exist, we must have $p \neq q$. WLOG, let $q > p$. We would like to *guarantee* a profit by betting on A and B on different websites with potentially different sizings. Let S_A, S_B be our sizings on A and B respectively. Notice that if we make bets where the probabilities total to more than 1, we are expected to lose money. Conversely, if our probabilities total to less than 1, we are expected to win money. (Exercise: verify this.)

Thus, we would like to bet on A on website 1 and B on website 2. This gives us the following equation,

$$\begin{aligned} S_A \cdot \frac{1-p}{p} - S_B &= S_B \cdot \frac{q}{1-q} - S_A \\ \frac{S_A}{p} &= \frac{S_B}{1-q} \\ S_A &= S_B \cdot \frac{p}{1-q} \end{aligned}$$

And our guaranteed winnings for some sizing S_B will be $S_B(\frac{1-p}{p} \cdot \frac{p}{1-q} - 1) = S_B \cdot \frac{q-p}{1-q}$.

For the example above, we can now easily calculate our winnings as $100 \cdot \frac{.5}{.25} = 200$.